

§11.5 - Alternating Series

Example #1 (Alternating Harmonic Series)

The sum

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

exists.

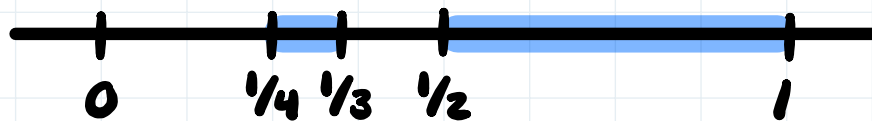
The Alternating Harmonic series

$$S = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}$$

satisfies

$$\begin{aligned} S_{2n} &= \sum_{k=1}^{2n} \frac{(-1)^{k+1}}{k} \\ &= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{2n-1} - \frac{1}{2n}\right) \\ &\leq 1 \end{aligned}$$

since the intervals are disjoint subsets of $(0, 1)$.



Thus S_{2n} is bounded and increasing and so it converges by the MCT.

Now

$$S_{2n+1} = S_{2n} + \frac{1}{2n+1}$$

and so

$$\begin{aligned}\lim_{n \rightarrow \infty} S_{2n+1} &= \lim_{n \rightarrow \infty} \left(S_{2n} + \frac{1}{2n+1} \right) \\ &= \lim_{n \rightarrow \infty} S_{2n} + 0 \\ &= \lim_{n \rightarrow \infty} S_{2n}\end{aligned}$$

since

$$\lim_{n \rightarrow \infty} \frac{1}{2n+1} = 0.$$

Thus the Alternating Harmonic series converges.

We can generalize the result.

Theorem #1 (Alternating Series Test)

Suppose (b_n) satisfies

$$\lim_{n \rightarrow \infty} b_n = 0 \quad \text{and} \quad b_{n+1} \leq b_n \text{ for all } n$$

then

$$\sum (-1)^n b_n$$

converges.

Remark

We will see later that

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = \ln(2).$$

Example #2

Determine if

$$\sum_{k=2}^{\infty} (-1)^k \frac{\ln(k)}{k^2}$$

converges.

Eventually

$$f(x) = \frac{\ln(x)}{x^2}$$

decreases to 0 since

$$f'(x) = \frac{1 - 2\ln(x)}{x^3} < 0$$

for $x > e^{1/2}$ and

$$\frac{\ln(x)}{x^2} \rightarrow 0$$

by L'Hospital.

Then

$$\sum_{k=2}^{\infty} (-1)^k \frac{\ln(k)}{k^2}$$

converges by the Alternating Series test.

Theorem #2 (Error Estimate for the AST)

Assume

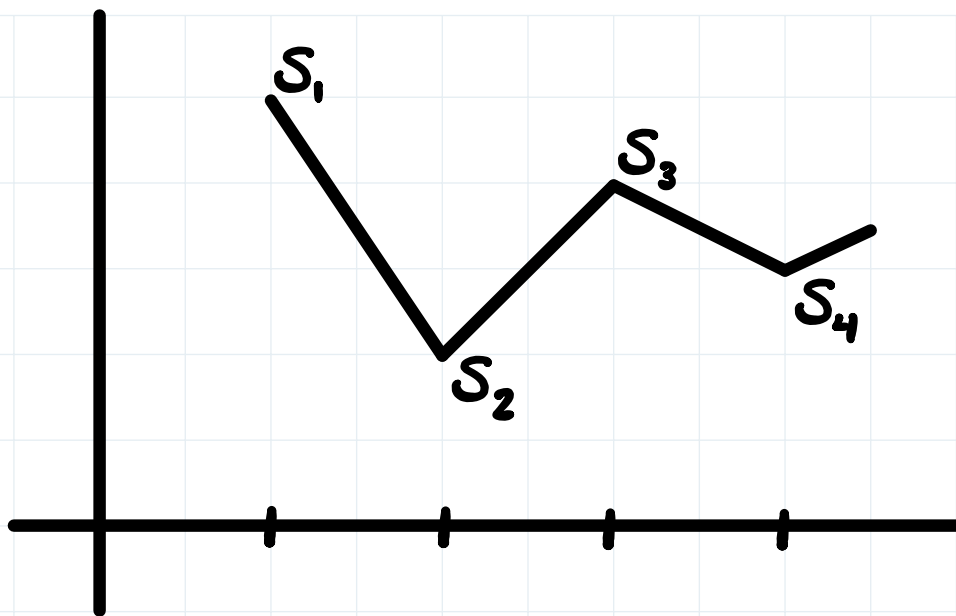
$$S = \sum_{k=1}^{\infty} (-1)^{k+1} b_k$$

converges by the Alternating Series Test.

Then

$$|S - S_n| \leq b_{n+1}$$

where S_n is the n th partial sum of S .



With (S_{2n}) increasing

$$S_{2n} \leq S$$

and by a similar argument (S_{2n+1}) is decreasing since (b_n) is decreasing to 0,

$$0 \leq b_{2n+3} \leq b_{2n+2}$$

$$0 \leq b_{2n+2} - b_{2n+3}$$

so

$$\begin{aligned} S_{2n+3} &= S_{2n+1} - b_{2n+2} + b_{2n+3} \\ &= S_{2n+1} - \underbrace{(b_{2n+2} - b_{2n+3})}_{\text{non negative}} \end{aligned}$$

and we find

$$S_{2n+3} \leq S_{2n+1}.$$

Take

$$S_{2n} \leq S \leq S_{2n+1}$$

and subtract S_{2n}

$$0 \leq S - S_{2n} \leq S_{2n+1} - S_{2n} = b_{2n+1}.$$

Do a similar thing with S_{2n+2} and S_{2n+1} :

$$S_{2n+2} \leq S \leq S_{2n+1}$$

but subtract S_{2n+1} ,

$$S_{2n+2} - S_{2n+1} \leq S - S_{2n+1} \leq 0.$$

This shows

$$|S - S_{2n+1}| \leq |S_{2n+2} - S_{2n+1}| = b_{2n+2}$$

Example #3

Choose $n \in \mathbf{N}$ such that

$$|S - S_n| \leq 10^{-4}$$

where

$$S = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2}.$$

Solve

$$b_{n+1} = \frac{1}{(n+1)^2} \leq 10^{-4}$$

$$10^4 \leq (n+1)^2$$

$$100 \leq n+1$$

so $n \geq 99$.

Then

$$S_{99} \approx 0.822518$$

and more advanced methods show

$$S = \frac{\pi^2}{12} \approx 0.822467.$$

Thus

$$|S_{99} - S| \approx 5.05 \cdot 10^{-5} < 10^{-4}.$$

Example #4

Compare

$$\ln(2) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \dots$$

to

$$B = 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \dots$$

What are the next terms in B ?

$$B = 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} \\ - \frac{1}{12} + \frac{1}{7} - \frac{1}{14} - \dots$$

In B we add an odd denominator term, then subtract two even denominator terms:

"+, -, -"

Note

$$B = \left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \dots \\ = \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \dots \\ = \frac{1}{2} \ln(2).$$

Yet both sums contain the exact same terms!
Here is what drives the spooky behavior:

1. $\sum (-1)^{n+1} \frac{1}{n}$ converges

2. $\sum \frac{1}{n}$ diverges